

PART 3: Gradient Descent Manual Calculation.

Given Information.

Linear Equation $\Rightarrow \hat{y} = mx + b$

Data points $\Rightarrow (1, 3)$ and $(3, 6)$.

Initial values $\Rightarrow m_0 = -1, b_0 = 1$

Learning rate $\Rightarrow \alpha = 0.1$

Number of data points $\Rightarrow n = 2$.

Cost function (MSE, using the common convention with factor $1/(2n)$):

$$J(m, b) = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y})^2,$$

$$\text{But } \hat{y} = mx + b$$

$$\Rightarrow J(m, b) = \frac{1}{2n} \sum_{i=1}^n (y_i - (mx + b))^2.$$

For $n = 2$, the prefactor is $\frac{1}{2n}$

$$\Rightarrow \frac{1}{2(2)} = \frac{1}{4} = 0.25.$$

Derive gradients.

$$\text{From } J(m, b) = \frac{1}{2n} \sum_{i=1}^n (y_i - (mx + b))^2$$

$$\text{Let } e_i = y_i - (mx + b).$$

$$\text{Cost/Error/Loss function } \Rightarrow J_{(m, b)} = \frac{1}{2n} \sum_{i=1}^n e_i^2$$

Partial derivative w.r.t m :

$$\Rightarrow \frac{\partial J}{\partial m} = \frac{\partial}{\partial m} \left(\frac{1}{2n} \sum_{i=1}^n e_i^2 \right)$$

$$\Rightarrow \frac{1}{2n} \sum_{i=1}^n 2e_i \cdot \frac{\partial e_i}{\partial m} \quad (\text{Chain rule})$$

$$\text{But } e_i = y_i - (mx + b) \Rightarrow \frac{\partial e_i}{\partial m} = -x,$$

$$\text{Now } \frac{\partial J}{\partial m} = \frac{1}{2n} \sum_{i=1}^n 2e_i(-x) \\ \Rightarrow -\frac{1}{n} \sum_{i=1}^n e_i x.$$

Partial derivative w.r.t b :

$$\frac{\partial J}{\partial b} = \frac{1}{2n} \sum_{i=1}^n 2e_i \cdot \frac{\partial e_i}{\partial b}$$

$$\text{and } \frac{\partial e_i}{\partial b} = -1. \quad \text{Hence}$$

$$\frac{\partial e_i}{\partial b} = -\frac{1}{n} \sum_{i=1}^n e_i.$$

So final gradient formulas:

$$\boxed{\frac{\partial J}{\partial m} = -\frac{1}{n} \sum_{i=1}^n x(y_i - (mx + b))}$$

$$\boxed{\frac{\partial J}{\partial b} = -\frac{1}{n} \sum_{i=1}^n (y_i - (mx + b))}$$

Update rules (gradient descent):

$$m_{\text{new}} = m_{\text{old}} - \alpha \frac{\partial J}{\partial m},$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \frac{\partial J}{\partial b}$$

Iteration

Iteration 1

Given Informations:

$m_0 = -1$, $b_0 = 1$, data points (1,3) and (3,6)

lets compute initial predictions:

$$\begin{aligned}\text{For } x_1 = 1: \hat{y}_1 &= m_0 x_1 + b_0 \\ &= (-1)(1) + 1 \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{For } x_2 = 3: \hat{y}_2 &= m_0 x_2 + b_0 \\ &= (-1)(3) + 1 \\ &= -2\end{aligned}$$

$$\text{Loss}_1 = y_1 - \hat{y}_1 = 3 - 0 = 3$$

$$\text{Loss}_2 = y_2 - \hat{y}_2 = 6 - (-2) = 8$$

$$\text{Compute cost} \Rightarrow J_0 = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

But $n=2$

$$J_0 = \frac{1}{2(2)} \cdot ((\text{Loss}_1)^2 + (\text{Loss}_2)^2)$$

$$\Rightarrow \frac{1}{4} \cdot (3^2 + 8^2)$$

$$= 0.25 \cdot (9 + 64)$$

$$= 0.25 \times 73$$

$$= 18.25$$

Now Gradient compute

$$\Rightarrow -\frac{1}{n} \sum_{i=1}^n e_i x = \frac{\partial J}{\partial m}, n=2$$

$$\text{So } \left. \frac{\partial J}{\partial m} \right|_0 = -\frac{1}{2} \cdot (e_1 x_1 + e_2 x_2)$$

But we have our $e_i = \text{Loss}$

$$\Rightarrow -\frac{1}{2} \cdot (\text{Loss}_1 x_1 + \text{Loss}_2 x_2)$$

$$= -0.5 \cdot (3(1) + 8(3))$$

$$= -0.5 \cdot (3 + 24)$$

$$= -13.5$$

We do same for $\left. \frac{\partial J}{\partial b} \right|_0$

$$\text{where } \frac{\partial J}{\partial b} = -\frac{1}{n} \sum_{i=1}^n e_i$$

And still here $e_i = \text{Loss}$

$$\Rightarrow -\frac{1}{2} (\text{Loss}_1 + \text{Loss}_2)$$

$$= -0.5 (3 + 8)$$

$$= -0.5 \times 11$$

$$= -5.5$$

What we do now is to update parameters:

$$* m_1 = m_0 - \alpha \cdot \left. \frac{\partial J}{\partial m} \right|_0$$

$$m_1 = -1 - 0.1 \cdot (-13.5)$$

$$m_1 = -1 + 1.35$$

$$m_1 = 0.35$$

$$* b_1 = b_0 - \alpha \cdot \left. \frac{\partial J}{\partial b} \right|_0$$

$$= 1 - 0.1 \cdot (-5.5)$$

$$= 1 + 0.55$$

$$\therefore b_1 = 1.55$$

So we now say after our first iteration:

$$m_1 = 0.35$$

$$b_1 = 1.55$$

$$J_0 = 18.25$$

Iteration 2

Given Information :

$$m_1 = 0.35,$$

$$b_1 = 1.55, \text{ data points } (1, 3) \text{ and } (3, 6)$$

So we now start with $\bar{e}$ predictions :

Now we have our equation

$$= \hat{y}_1 = m_1 x_1 + b_1$$

$$\Rightarrow \hat{y}_1 = 0.35(1) + 1.55$$
$$= 0.35 + 1.55$$
$$= 1.90$$

$$\text{Now } \hat{y}_2 = m_1 x_2 + b_1$$

$$\hat{y}_2 = 0.35(3) + 1.55$$
$$\Rightarrow \hat{y}_2 = 1.05 + 1.55$$
$$= 2.60$$

And we find our new errors again with new values.

$$\text{error}_1 = y_1 - \hat{y}_1$$
$$= 3 - 1.90$$
$$= 1.10$$

$$\text{error}_2 = 6 - 2.60$$
$$= 3.40$$

Cost J_1 (after first update, compute cost using these errors):

$$J_1 = 0.25 \cdot (1.10^2 + 3.40^2)$$

from $\bar{e}$ formula $\frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

where $n=2$

$$J_1 = 0.25(1.21 + 11.56)$$
$$= 0.25(12.77)$$
$$= 3.19$$

Gradient updates:

$$\frac{\partial J}{\partial m_1} = -\frac{1}{n} \sum_{i=1}^n e_i x_i$$

now to $e_i = \text{error}_i, n=2$

$$\frac{\partial J}{\partial m_1} = -\frac{1}{2} (1.10(1) + 3.40(3))$$
$$= -0.5(1.10 + 10.20)$$
$$= -0.5 \times 11.30$$
$$= -5.65$$

$$\text{Same for } \frac{\partial J}{\partial b_1} = -\frac{1}{n} \sum_{i=1}^n e_i$$

Here to $e_i = \text{error}_i$, which is new error.

$$\frac{\partial J}{\partial b_1} = -\frac{1}{2} (1.10 + 3.40)$$
$$= -0.5(4.50)$$
$$= -2.25$$

Updates:

$$m_2 = m_1 - \alpha \frac{\partial J}{\partial m_1}$$

$$m_2 = 0.35 - 0.1(-5.65)$$
$$= 0.35 + 0.565$$
$$= 0.915$$

Updates :

$$b_2 = b_1 - \alpha \cdot \left. \frac{\partial J}{\partial b_1} \right|_1$$

where $b_1 = 1.55$, $\alpha = 0.1$

$$\text{and } \left. \frac{\partial J}{\partial b_1} \right|_1 = -2.25$$

$$\Rightarrow b_2 = 1.55 - 0.1(-2.25) \\ = 1.55 + 0.225$$

$$\Rightarrow b_2 = 1.775$$

After iteration 2 :

$$m_2 = 0.92 \text{ (2d.p)}$$

$$b_2 = 1.78 \text{ (2d.p)}$$

$$J_1 = 3.19 \text{ (2d.p)}$$

Observed trend & interpretation

Cost (J) decreased strongly:

$$* J_0 = 18.25 \rightarrow 3.19$$

And when the third iteration is done the cost ~~auth~~ would towards zero (0).

So we can say the model is reducing the mean squared error each iteration.

Parameters:

m moved from $-1.00 \rightarrow 0.35$

$\rightarrow 0.92$

and,

b moved from $1.00 \rightarrow 1.55$

$\rightarrow 1.78$

* Are they moving toward a true line
a exact line passing through a two points
(1,3) and (3,6) has slope

$$m^* = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 3}{3 - 1} = \frac{3}{2} = 1.5,$$

and a intercept

$$\begin{aligned} b^* &= y_1 - m^* \cdot x_1 \\ &= 3 - (1.5 \times 1) \\ &= 1.5. \end{aligned}$$

Our parameters after 2 updates ($m_2 = 0.92$, $b_2 = 1.78$) are moving toward that true solution (slope 1.5, intercept 1.5).

The slope is increasing toward 1.5;

the intercept somehow overshoots above 1.5 but will balance as more iterations proceed.

(gradient descent moves both to reduce total error). The cost is sharply decreasing, so yes - error is being reduced. $\approx$